

# EXTENDED STAR

## 1. Project Overview

In this documentation, the design of the extended star topology is addressed together with its configs and implementation all done using Cisco Packet Tracer. In this project setup all the departments were assigned different VLAN and all integrated with the use of distribution, core switch and the router.

- Project Title: ExtendedStar\_PathFinder
- Objective: Demonstrate how data moves in an extended star topology which is a variant version of the star topology.
- Topology: Extended Star Topology
- Core Technology: VLAN with segmentation

## 2. Materials used:

| Item        | Quantity | Model     |
|-------------|----------|-----------|
| Router      | 1        | 2901      |
| PC          | 10       | PC-PT     |
| Switch      | 5        | 2960-24TT |
| Core Switch | 1        | 3560-24PS |

## 3. Network design and Topology:

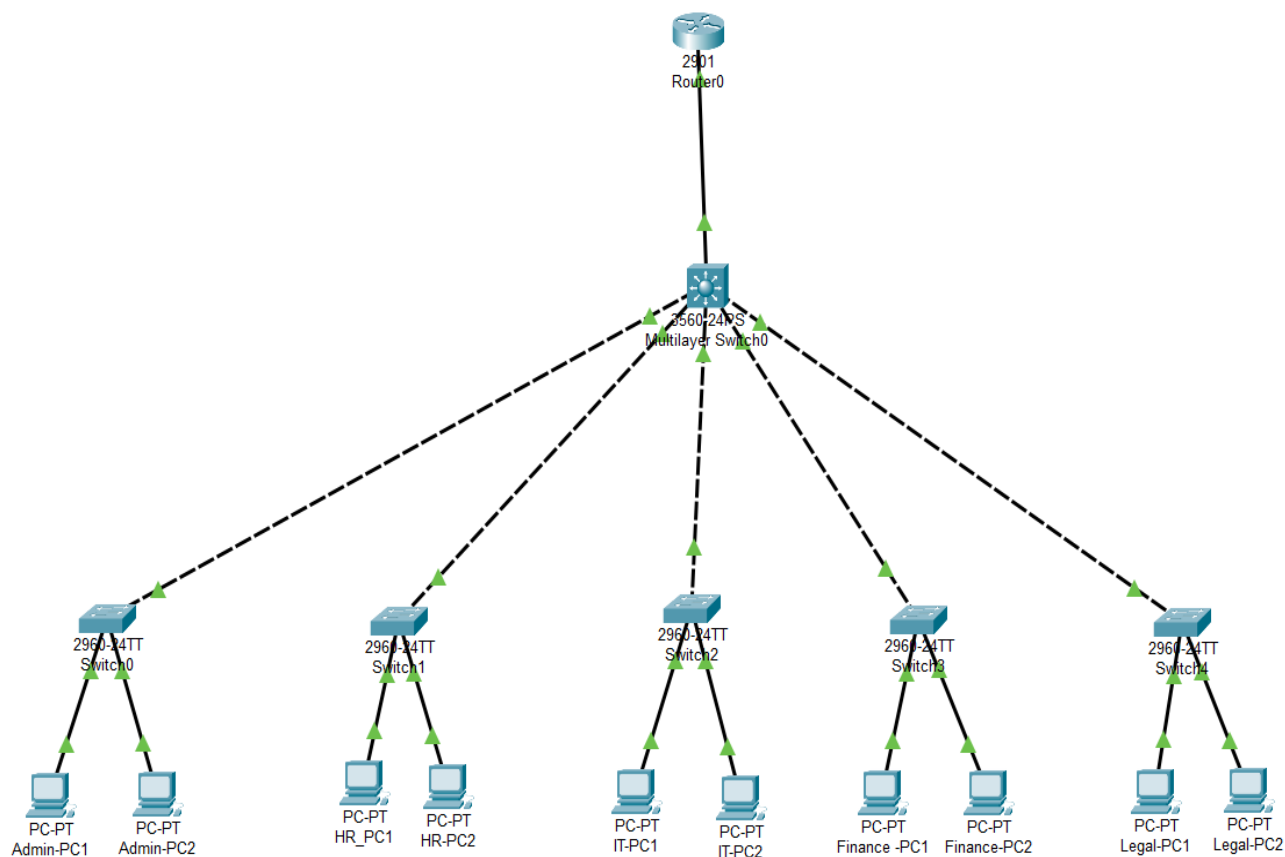
Physical Topology: Hierarchical star

- Core switch: One multilayered switch handling the routing
- Router: One router which facilitates communication between the system and the external environment.
- Access switches: 5 connected directly to the core switches
- End devices: 10 connected to the access switches.

Logical design with VLAN segmentation

Each access switch is able to handle different VLANS because IP routing is done by the multilayered core to which they are all connected. The router

itself deals with only facilitating inside and outside the working network.



#### Diagram Explanation:

The entire set up is a hierarchical setup with the router connected to the core and the core connecting up to 5 distribution switches, the access switches further each connects two end devices. In this diagram, the multilayered switch acts as the router and does the inter-VLAN routing. The end devices send any data/message to the access switch which is then sent to the core, and it is there where the core decides whether routing is needed or not. If it's needed the core forwards it to the relevant department otherwise the access switch will forward the packet. Whenever data needs to be sent out of the system the core switch sends it to the router where it will be verified and sent externally, this can be to the internet or otherwise.

#### 4. Configurations:

Router config

```
Router> enable
```

```
Router# configure terminal
```

```
Router(config)# hostname R-CORE
```

```
! Configure physical interface
```

```
R-CORE(config)# interface gigabitEthernet 0/1
```

```
R-CORE(config-if)# no shutdown
```

```
R-CORE(config-if)# exit
```

```
! Subinterface for Admin VLAN 10
```

```
R-CORE(config)# interface gigabitEthernet 0/1.10
```

```
R-CORE(config-if)# encapsulation dot1Q 10
```

```
R-CORE(config-if)# ip address 192.168.10.254  
255.255.255.0
```

```
R-CORE(config-if)# exit
```

```
! Subinterface for HR VLAN 20
```

```
R-CORE(config)# interface gigabitEthernet 0/1.20
```

```
R-CORE(config-if)# encapsulation dot1Q 20
```

```
R-CORE(config-if)# ip address 192.168.20.254  
255.255.255.0
```

```
R-CORE(config-if)# exit
```

```
! Subinterface for IT VLAN 30
```

```
R-CORE(config)# interface gigabitEthernet 0/1.30
```

```
R-CORE(config-if)# encapsulation dot1Q 30
```

```
R-CORE(config-if)# ip address 192.168.30.254  
255.255.255.0
```

```
R-CORE(config-if)# exit
```

```
! Subinterface for Finance VLAN 40
```

```
R-CORE(config)# interface gigabitEthernet 0/1.40
```

```
R-CORE(config-if)# encapsulation dot1Q 40
```

```
R-CORE(config-if)# ip address 192.168.40.254  
255.255.255.0
```

```
R-CORE(config-if)# exit
```

```
! Subinterface for Legal VLAN 50
R-CORE(config)# interface gigabitEthernet 0/1.50
R-CORE(config-if)# encapsulation dot1Q 50
R-CORE(config-if)# ip address 192.168.50.254
255.255.255.0
R-CORE(config-if)# exit
R-CORE(config)# ip routing
```

```
! Save configuration
R-CORE(config)# exit
R-CORE# copy running-config startup-config
```

#### b) CORE ACCESS SWITCH

```
CoreSwitch> enable
CoreSwitch# configure terminal
CoreSwitch(config)# hostname SW-CORE
```

```
! Create VLANs 10-50
SW-CORE(config)# vlan 10
SW-CORE(config-vlan)# name ADMIN
SW-CORE(config-vlan)# exit
```

```
SW-CORE(config)# vlan 20
SW-CORE(config-vlan)# name HR
SW-CORE(config-vlan)# exit
```

```
SW-CORE(config)# vlan 30
SW-CORE(config-vlan)# name IT
SW-CORE(config-vlan)# exit
```

```
SW-CORE(config)# vlan 40
SW-CORE(config-vlan)# name FINANCE
SW-CORE(config-vlan)# exit
```

```
SW-CORE(config)# vlan 50
```

```
SW-CORE(config-vlan)# name LEGAL
SW-CORE(config-vlan)# exit
```

! Configure SVI interfaces for each VLAN

```
SW-CORE(config)# interface vlan 10
SW-CORE(config-if)# ip address 192.168.10.1
255.255.255.0
SW-CORE(config-if)# no shutdown
SW-CORE(config-if)# exit
```

```
SW-CORE(config)# interface vlan 20
SW-CORE(config-if)# ip address 192.168.20.1
255.255.255.0
SW-CORE(config-if)# no shutdown
SW-CORE(config-if)# exit
```

```
SW-CORE(config)# interface vlan 30
SW-CORE(config-if)# ip address 192.168.30.1
255.255.255.0
SW-CORE(config-if)# no shutdown
SW-CORE(config-if)# exit
```

```
SW-CORE(config)# interface vlan 40
SW-CORE(config-if)# ip address 192.168.40.1
255.255.255.0
SW-CORE(config-if)# no shutdown
SW-CORE(config-if)# exit
```

```
SW-CORE(config)# interface vlan 50
SW-CORE(config-if)# ip address 192.168.50.1
255.255.255.0
SW-CORE(config-if)# no shutdown
SW-CORE(config-if)# exit
```

! Enable routing

```
SW-CORE(config)# ip routing
```

```
! Configure trunk ports to access switches (Fa0/1-5)
```

```
SW-CORE(config)# interface range fastEthernet 0/1 - 5
```

```
SW-CORE(config-if-range)# switchport mode trunk
```

```
SW-CORE(config-if-range)# switchport trunk encapsulation dot1q
```

```
SW-CORE(config-if-range)# switchport trunk allowed vlan 10,20,30,40,50
```

```
SW-CORE(config-if-range)# no shutdown
```

```
SW-CORE(config-if-range)# exit
```

```
! Save configuration
```

```
SW-CORE(config)# exit
```

```
SW-CORE# copy running-config startup-config
```

#### c) ADMIN DEPARTMENT

```
Switch> enable
```

```
Switch# configure terminal
```

```
Switch(config)# hostname SW-ADMIN
```

```
! Create VLAN 10
```

```
SW-ADMIN(config)# vlan 10
```

```
SW-ADMIN(config-vlan)# name ADMIN
```

```
SW-ADMIN(config-vlan)# exit
```

```
! Configure access ports for PCs (Fa0/1-2)
```

```
SW-ADMIN(config)# interface range fastEthernet 0/1-2
```

```
SW-ADMIN(config-if-range)#
```

```
switchport mode access
```

```
SW-ADMIN(config-if-range)#  
switchport access vlan 10  
SW-ADMIN(config-if-range)# spanning-  
tree portfast  
SW-ADMIN(config-if-range)# no  
shutdown  
SW-ADMIN(config-if-range)# exit
```

```
! Configure trunk port to Core switch  
(Fa0/3)  
SW-ADMIN(config)# interface  
fastEthernet 0/3  
SW-ADMIN(config-if)# switchport  
mode trunk  
SW-ADMIN(config-if)# switchport trunk  
native vlan 999  
SW-ADMIN(config-if)# switchport trunk  
allowed vlan 10,20,30,40,50  
SW-ADMIN(config-if)# no shutdown  
SW-ADMIN(config-if)# exit
```

```
! Save configuration  
SW-ADMIN(config)# exit  
SW-ADMIN# copy running-config  
startup-config
```

#### d) HR DEPARTMENT

```
Switch> enable  
Switch# configure terminal  
Switch(config)# hostname SW-HR
```

```
! Create VLAN 20  
SW-HR(config)# vlan 20  
SW-HR(config-vlan)# name HR  
SW-HR(config-vlan)# exit
```

```
! Configure access ports for PCs (Fa0/1-2)
SW-HR(config)# interface range fastEthernet 0/1-2
SW-HR(config-if-range)# switchport mode access
SW-HR(config-if-range)# switchport access vlan 20
SW-HR(config-if-range)# spanning-tree portfast
SW-HR(config-if-range)# no shutdown
SW-HR(config-if-range)# exit
```

```
! Configure trunk port to Core switch (Fa0/3)
SW-HR(config)# interface fastEthernet 0/3
SW-HR(config-if)# switchport mode trunk
SW-HR(config-if)# switchport trunk native vlan 999
SW-HR(config-if)# switchport trunk allowed vlan
10,20,30,40,50
SW-HR(config-if)# no shutdown
SW-HR(config-if)# exit
```

```
! Save configuration
SW-HR(config)# exit
SW-HR# copy running-config startup-config
```

e) IT DEPARTMENT

```
Switch> enable
Switch# configure terminal
Switch(config)# hostname SW-IT
```

```
! Create VLAN 30
SW-IT(config)# vlan 30
SW-IT(config-vlan)# name IT
SW-IT(config-vlan)# exit
```

```
! Configure access ports for PCs (Fa0/1-2)
SW-IT(config)# interface range fastEthernet 0/1-2
SW-IT(config-if-range)# switchport mode access
```



```
SW-IT(config-if-range)# switchport access vlan 30
SW-IT(config-if-range)# spanning-tree portfast
SW-IT(config-if-range)# no shutdown
SW-IT(config-if-range)# exit
```

! Configure trunk port to Core switch (Fa0/3)

```
SW-IT(config)# interface fastEthernet 0/3
SW-IT(config-if)# switchport mode trunk
SW-IT(config-if)# switchport trunk native vlan 999
SW-IT(config-if)# switchport trunk allowed vlan
10,20,30,40,50
SW-IT(config-if)# no shutdown
SW-IT(config-if)# exit
```

! Save configuration

```
SW-IT(config)# exit
SW-IT# copy running-config startup-config
```

#### f) FINANCE DEPARTMENT

```
Switch> enable
Switch# configure terminal
Switch(config)# hostname SW-FINANCE
```

! Create VLAN 40

```
SW-FINANCE(config)# vlan 40
SW-FINANCE(config-vlan)# name FINANCE
SW-FINANCE(config-vlan)# exit
```

! Configure access ports for PCs (Fa0/1-2)

```
SW-FINANCE(config)# interface range fastEthernet 0/1-2
SW-FINANCE(config-if-range)# switchport mode access
SW-FINANCE(config-if-range)# switchport access vlan
40
SW-FINANCE(config-if-range)# spanning-tree portfast
SW-FINANCE(config-if-range)# no shutdown
```

```
SW-FINANCE(config-if-range)# exit
```

```
! Configure trunk port to Core switch (Fa0/3)
```

```
SW-FINANCE(config)# interface fastEthernet 0/3
```

```
SW-FINANCE(config-if)# switchport mode trunk
```

```
SW-FINANCE(config-if)# switchport trunk native vlan 999
```

```
SW-FINANCE(config-if)# switchport trunk allowed vlan  
10,20,30,40,50
```

```
SW-FINANCE(config-if)# no shutdown
```

```
SW-FINANCE(config-if)# exit
```

```
! Save configuration
```

```
SW-FINANCE(config)# exit
```

```
SW-FINANCE# copy running-config startup-config
```

#### g) LEGAL DEPARTMENT

```
Switch> enable
```

```
Switch# configure terminal
```

```
Switch(config)# hostname SW-LEGAL
```

```
! Create VLAN 50
```

```
SW-LEGAL(config)# vlan 50
```

```
SW-LEGAL(config-vlan)# name LEGAL
```

```
SW-LEGAL(config-vlan)# exit
```

```
! Configure access ports for PCs (Fa0/1-2)
```

```
SW-LEGAL(config)# interface range fastEthernet 0/1-2
```

```
SW-LEGAL(config-if-range)# switchport mode access
```

```
SW-LEGAL(config-if-range)# switchport access vlan 50
```

```
SW-LEGAL(config-if-range)# spanning-tree portfast
```

```
SW-LEGAL(config-if-range)# no shutdown
```

```
SW-LEGAL(config-if-range)# exit
```

```
! Configure trunk port to Core switch (Fa0/3)
```

```
SW-LEGAL(config)# interface fastEthernet 0/3
```

```
SW-LEGAL(config-if)# switchport mode trunk
SW-LEGAL(config-if)# switchport trunk native vlan 999
SW-LEGAL(config-if)# switchport trunk allowed vlan
10,20,30,40,50
SW-LEGAL(config-if)# no shutdown
SW-LEGAL(config-if)# exit
```

! Save configuration

```
SW-LEGAL(config)# exit
```

```
SW-LEGAL# copy running-config startup-config
```

## 5. IP ADDRESSING:

The departments are all scattered along different VLANS, and the end devices are configured statically as per the relevant dept using IPv4 addressing only.

- VLAN 10 – Admin Department
  - ◆ Network address: 192.168.10.0/24
  - ◆ Subnet Mask: 255.255.255.0
  - ◆ Default gateway: 192.168.10.1
- VLAN 20- HR Department
  - Network Address:192.168.20.0/24
  - Subnet mask:255.255.255.0
  - Default gateway: 192.168.20.1
- VLAN 30- IT Department
  - Network Address: 192.168.30.0/24
  - Subnet mask: 255.255.255.0
  - Default gateway: 192.168.30.1

- VLAN 40- Finance Department
  - Network Address: 192.168.40.0/24
  - Subnet mask: 255.255.255.0
  - Default gateway: 192.168.40.1
  
- VLAN 50- Legal Department
  - Network Address: 192.168.50.0/24
  - Subnet Mask: 255.255.255.0
  - Default gateway: 192.168.50.1

All the devices on the network are connected using cables/wires.

| Device      | IP address   | VLAN | Remark                          |
|-------------|--------------|------|---------------------------------|
| Admin-PC1   | 192.168.10.2 | 10   | Connected to the admin switch   |
| Admin-PC2   | 192.168.10.3 | 10   | Connected to admin switch       |
| HR-PC1      | 192.168.20.2 | 20   | Connected to the HR switch      |
| HR-PC2      | 192.168.20.3 | 20   | Connected to HR switch          |
| IT-PC1      | 192.168.30.2 | 30   | Connected to IT switch          |
| IT-PC2      | 192.168.30.3 | 30   | Connected to IT switch          |
| Finance-PC1 |              | 40   | Connected to the Finance switch |
| Finance-PC2 |              | 40   | Connected to the finance switch |
| Legal-PC1   | 192.168.50.3 | 50   | Connected to the legal switch   |
| Legal-PC2   | 192.168.50.2 | 50   | Connected to legal switch       |

## 6. TESTING AND VERIFICATION:

- a) The cdp neighbor as seen by the core switch

```

Router#show cdp neighbor
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID        Local Intrfce   Holdtme    Capability   Platform    Port ID
Switch           Gig 0/0                168
Router#

```

b) The cdp neighbor as seen by the core switch

```

Switch#show cdp neighbor
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID        Local Intrfce   Holdtme    Capability   Platform    Port ID
Router           Gig 0/1                158        R           C2900       Gig 0/0
ITSwitch         Fas 0/3                158        S           2960        Fas 0/3
AdminSwitch      Fas 0/1                158        S           2960        Fas 0/3
HRSwitch         Fas 0/2                158        S           2960        Fas 0/3
FinanceSwitch
                 Fas 0/4                158        S           2960        Fas 0/3
LegalSwictch     Fas 0/5                158        S           2960        Fas 0/3

```

c) The IP interfaces on the core switch

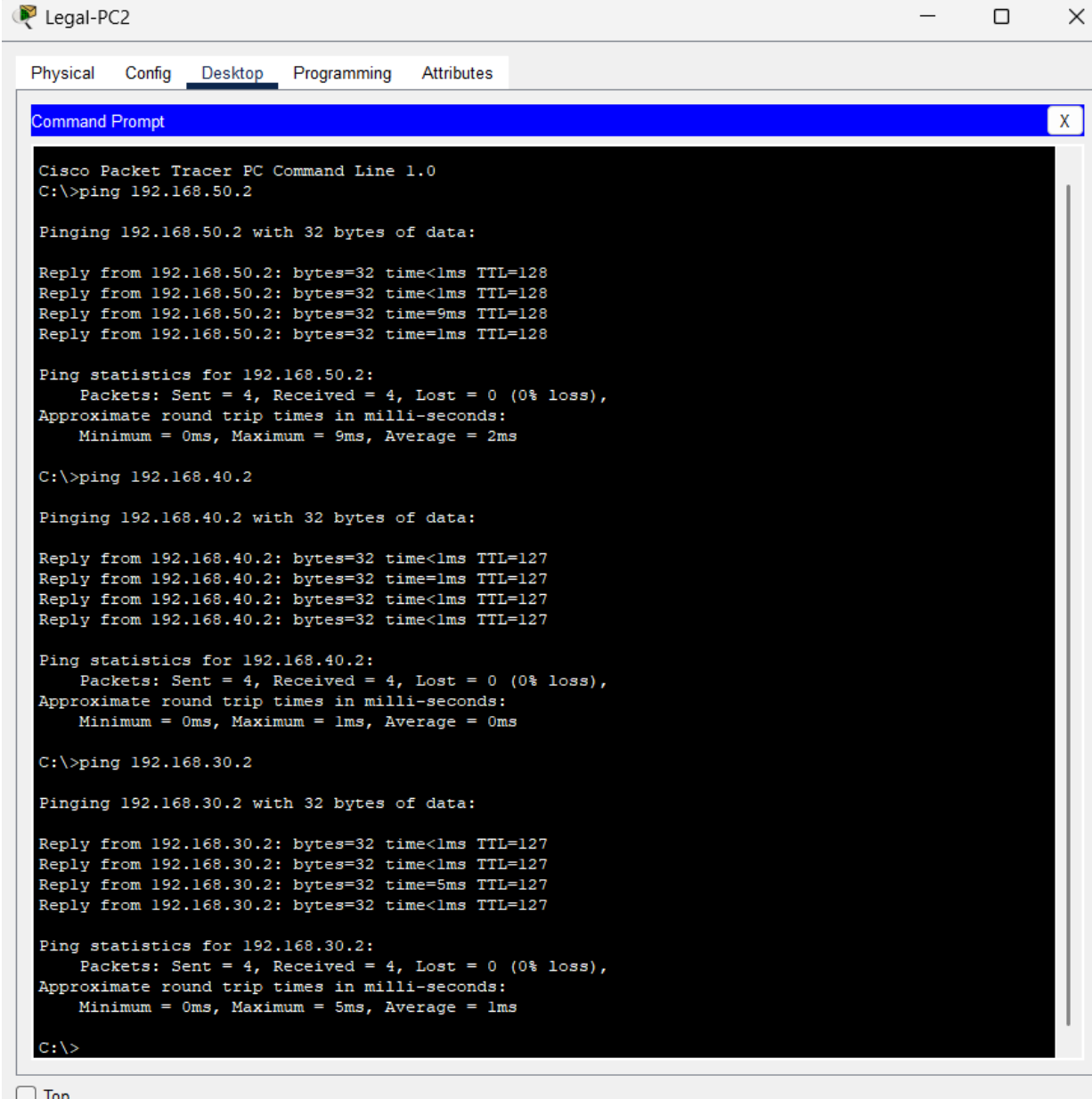
```

SW-CORE#show ip interface brief
Interface        IP-Address      OK? Method Status      Protocol
FastEthernet0/1  unassigned      YES unset  up          up
FastEthernet0/2  unassigned      YES unset  up          up
FastEthernet0/3  unassigned      YES unset  up          up
FastEthernet0/4  unassigned      YES unset  up          up
FastEthernet0/5  unassigned      YES unset  up          up
FastEthernet0/6  unassigned      YES unset  down        down
FastEthernet0/7  unassigned      YES unset  down        down
FastEthernet0/8  unassigned      YES unset  down        down
FastEthernet0/9  unassigned      YES unset  down        down
FastEthernet0/10 unassigned      YES unset  down        down
FastEthernet0/11 unassigned      YES unset  down        down
FastEthernet0/12 unassigned      YES unset  down        down
FastEthernet0/13 unassigned      YES unset  down        down
FastEthernet0/14 unassigned      YES unset  down        down
FastEthernet0/15 unassigned      YES unset  down        down
FastEthernet0/16 unassigned      YES unset  down        down
FastEthernet0/17 unassigned      YES unset  down        down
FastEthernet0/18 unassigned      YES unset  down        down
FastEthernet0/19 unassigned      YES unset  down        down
FastEthernet0/20 unassigned      YES unset  down        down
FastEthernet0/21 unassigned      YES unset  down        down
FastEthernet0/22 unassigned      YES unset  down        down
FastEthernet0/23 unassigned      YES unset  down        down
FastEthernet0/24 unassigned      YES unset  down        down
GigabitEthernet0/1 unassigned      YES unset  up          up
GigabitEthernet0/2 192.168.100.2   YES manual administratively down down
Vlan1            unassigned      YES unset  up          up
Vlan10           192.168.10.1    YES manual up          up
Vlan20           192.168.20.1    YES manual up          up
Vlan30           192.168.30.1    YES manual up          up
Vlan40           192.168.40.1    YES manual up          up
Vlan50           192.168.50.1    YES manual up          up
SW-CORE#

```

d) Ping within the same VLAN and different VLAN

In the output below the Legal department computer pinged within and outside its own VLAN



```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.50.2

Pinging 192.168.50.2 with 32 bytes of data:

Reply from 192.168.50.2: bytes=32 time<1ms TTL=128
Reply from 192.168.50.2: bytes=32 time<1ms TTL=128
Reply from 192.168.50.2: bytes=32 time=9ms TTL=128
Reply from 192.168.50.2: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.50.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 9ms, Average = 2ms

C:\>ping 192.168.40.2

Pinging 192.168.40.2 with 32 bytes of data:

Reply from 192.168.40.2: bytes=32 time<1ms TTL=127
Reply from 192.168.40.2: bytes=32 time=1ms TTL=127
Reply from 192.168.40.2: bytes=32 time<1ms TTL=127
Reply from 192.168.40.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.40.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.30.2

Pinging 192.168.30.2 with 32 bytes of data:

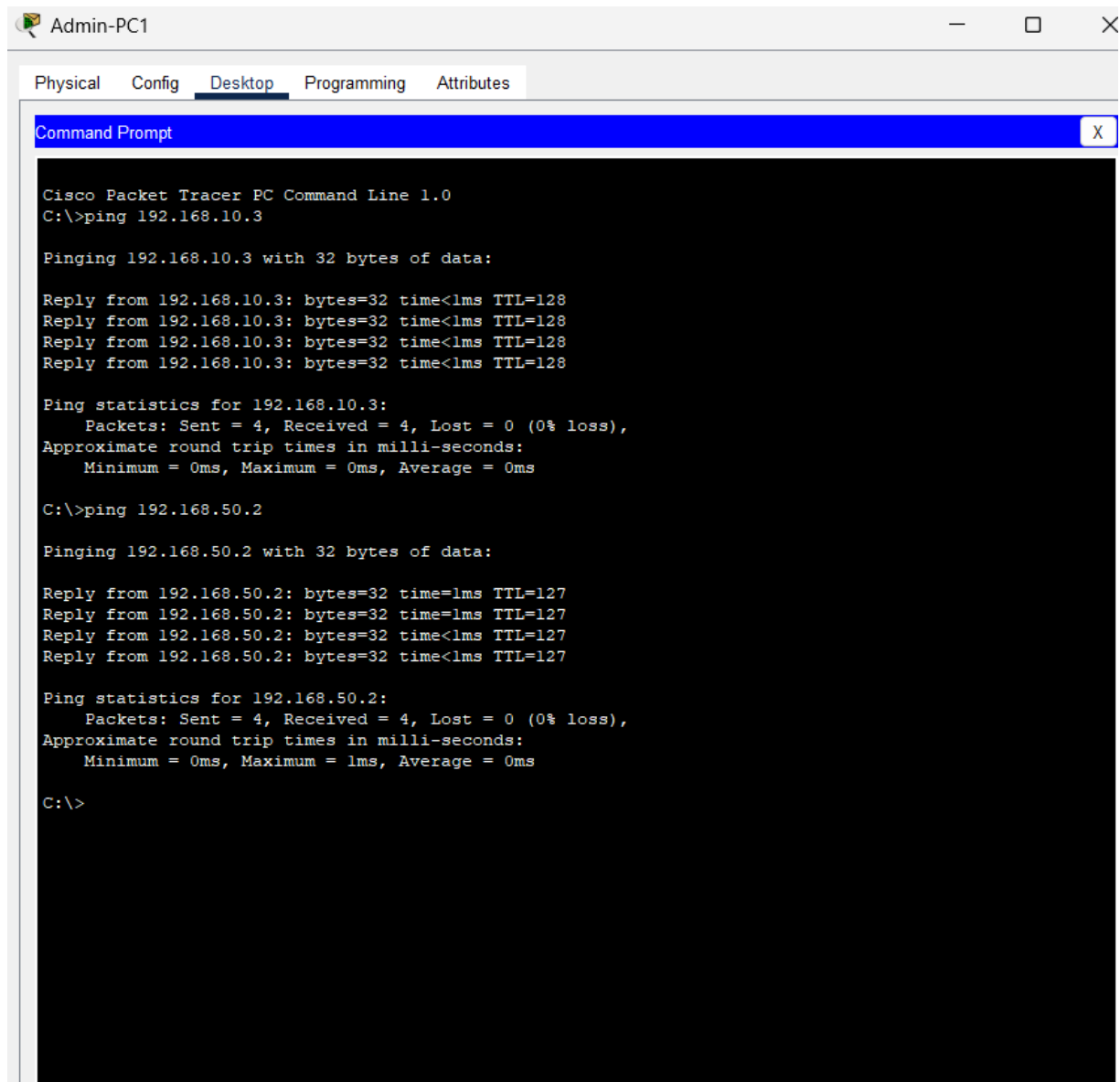
Reply from 192.168.30.2: bytes=32 time<1ms TTL=127
Reply from 192.168.30.2: bytes=32 time<1ms TTL=127
Reply from 192.168.30.2: bytes=32 time=5ms TTL=127
Reply from 192.168.30.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.30.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 5ms, Average = 1ms

C:\>
```

☐ Top

The diagram below shows the output given by a computer in the admin department within and outside its VLAN.



```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.3

Pinging 192.168.10.3 with 32 bytes of data:

Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.50.2

Pinging 192.168.50.2 with 32 bytes of data:

Reply from 192.168.50.2: bytes=32 time=1ms TTL=127
Reply from 192.168.50.2: bytes=32 time=1ms TTL=127
Reply from 192.168.50.2: bytes=32 time<1ms TTL=127
Reply from 192.168.50.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.50.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Using both diagrams it can be safely concluded that inter-VLAN does work within the network system and information is able to pass through as needed.

The diagram below shows the output from the router, and this indicates that there is communication between the router and all the devices on the system, therefore, should there be external communication needed the router will take care of it

```

Router#ping 192.168.100.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.100.2, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

Router#ping 192.168.20.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.20.2, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms

Router#ping 192.168.30.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.30.2, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/2/7 ms

Router#ping 192.168.40.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.40.2, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

Router#ping 192.168.50.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.50.2, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

```

The core switch can also ping the router suggesting that there is connection established between the two

```

SW-CORE#ping 192.168.100.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.100.1, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

```

## 7. Advantages:

- Inter-Vlan routing is fast as its directed by the MLS internally rather than the router.
- The segmented VLANS ensures that traffic flow of each department is contained in its own unless otherwise.
- The broadcasting issue is greatly reduced.
- The MLS gives the chance to be flexible, it can be used as a layer 3 switch or layer 2 as needed.

## 8. Disadvantages:



- The MLS controls a lot of things and if it goes down, the entire system is at risk.
- All traffic is directed to the MLS if it is compromised, the whole company is at risk.
- Troubleshooting can get complex as there are many things to check from.
- Hardware can be costly for this type of set up should the company expand.

## 9. Conclusion:

The network setup having a multilayered switch at the center that is connected to a router, and multiple access switches provides us with an efficient and structured way in which data is secured and able to move as needed. The placement of the router at the end of the topology ensures that the router is the one that first gets to deal with the external traffic can be directed internally thus providing the system with a bit of security. The VLAN segmentation further ensures security safety is maintained and enhanced, it also helps with logical separation and less broadcasting of information as some may even be sensitive. However, despite the core switch facilitating the IP routing, if it is compromised or fails, the entire set up is down. Despite all this the system works as intended as was shown by the outputs, thus making the system reliable, efficient, secure and perfect for a small enterprise such as SecureTrust Bank.